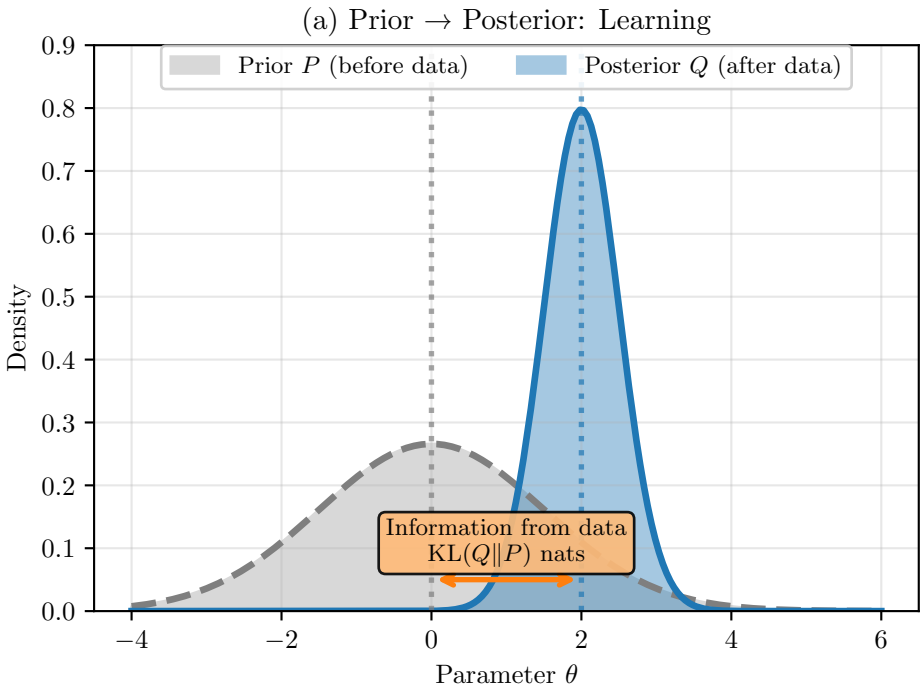
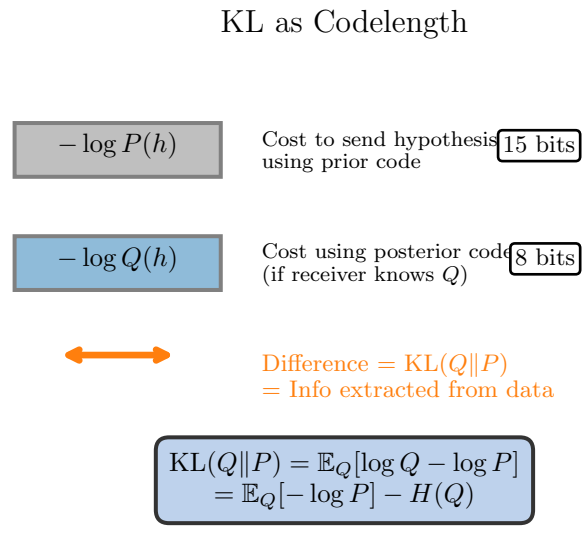

(a) Prior → Posterior: Learning
Prior P (before data)
Posterior Q (after data)
Density
Parameter θ
Information from data
KL($Q\|P$) nats
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
−4
−2
0
2
4
6


KL as Codelength
$-\log P(h)$
Cost to send hypothesis using prior code
15 bits
$-\log Q(h)$
Cost using posterior code (if receiver knows Q)
8 bits
Difference = KL($Q\|P$)
= Info extracted from data
KL($Q\|P$) = $\mathbb{E}_Q[\log Q - \log P]$
= $\mathbb{E}_Q[-\log P] - H(Q)$